Supplementary Material: Extending the MESA Approach for Solving General Mathematical Programming Models

1 Extending the MESA Approach for Solving General Mathematical Programming Models

As discussed in Section 4.2 of the main manuscript, the MESA method proposed in this study is not limited to approximating and accelerating ride-pooling problems. It can also be extended into a more general mathematical programming (MP) approximation framework aimed at reducing model size and simplifying problem structure across a broad class of optimization formulations. This supplementary note describes how the method can be generalized to broader MP models.

Consider the following general MP model:

$$\min_{\mathbf{x}\in\mathbb{R}^n,\,\mathbf{y}\in\mathbb{R}^m,\,\mathbf{z}\in\mathbb{R}^k} \quad C(\mathbf{x}) + H(\mathbf{y}) + P(\mathbf{z}) \tag{1}$$

$$\begin{aligned} \text{s.t.} \quad & A(\mathbf{x}) \le \mathbf{b}, \\ & D(\mathbf{x}) + E(\mathbf{y}) \le \mathbf{q}, \\ & G(\mathbf{y}) + S(\mathbf{z}) \le \mathbf{r}, \\ & \mathbf{x} \in \mathcal{X}, \quad \mathbf{y} \in \mathcal{Y}, \quad \mathbf{z} \in \mathcal{Z}. \end{aligned} \tag{2}$$

Here,

$$\begin{aligned} & C : \mathbb{R}^n \to \mathbb{R}, \qquad H : \mathbb{R}^m \to \mathbb{R}, \qquad P : \mathbb{R}^k \to \mathbb{R}, \\ & A : \mathbb{R}^n \to \mathbb{R}^{d_b}, \qquad D : \mathbb{R}^n \to \mathbb{R}^p, \qquad E : \mathbb{R}^m \to \mathbb{R}^p, \\ & G : \mathbb{R}^m \to \mathbb{R}^u, \qquad S : \mathbb{R}^k \to \mathbb{R}^u, \end{aligned}$$

with $\mathbf{b} \in \mathbb{R}^{d_b}$, $\mathbf{q} \in \mathbb{R}^p$, $\mathbf{r} \in \mathbb{R}^u$, $\mathcal{X} \subseteq \mathbb{R}^n$, $\mathcal{Y} \subseteq \mathbb{R}^m$, and $\mathcal{Z} \subseteq \mathbb{R}^k$.

Suppose that $\mathbf{y}$ and $\mathbf{z}$ are the primary decision variables, while $\mathbf{x}$ represents secondary decisions whose detailed optimization structure is computationally burdensome. We isolate the substructure

related to $\mathbf{x}$ and define, for any fixed feasible $\bar{\mathbf{y}}$, the value function

$$
\begin{aligned}
Q(\bar{\mathbf{y}}) = \min_{\mathbf{x}\in\mathbb{R}^n} \quad & C(\mathbf{x}) && (3)\\
\text{s.t.} \quad & A(\mathbf{x}) \le \mathbf{b}, \\
& D(\mathbf{x}) \le \mathbf{q} - E(\bar{\mathbf{y}}), \\
& \mathbf{x} \in \mathcal{X}. && (4)
\end{aligned}
$$

The subproblem takes $\bar{\mathbf{y}}$ as input and returns the corresponding optimal value $Q(\bar{\mathbf{y}})$. Directly solving this subproblem repeatedly inside the original MP model can be computationally expensive.

To address this challenge, MESA adopts a data-driven approximation. A predictor $F(\mathbf{y})$ is trained to approximate the mapping from $\mathbf{y}$ to the subproblem value $Q(\mathbf{y})$. The predictor may be linear (e.g., linear regression) or nonlinear (e.g., nonlinear regression, decision trees, random forests, or XGBoost). To permit direct integration into the remaining optimization model, the predictor must admit an MP representation, either exactly or through a suitable approximation.

The notion of a *predictor* here is intentionally abstract. In a specific application, its functional form and features should be customized to the underlying problem structure and operational setting. In the ride-pooling application studied in the main manuscript, the corresponding predictors estimate tour length, average pickup time, and average detour ratio.

One practical strategy is to construct a collection of MP-representable features derived from $\mathbf{y}$. Let

$$
\Theta(\mathbf{y}) = \big(g_1(\mathbf{y}), g_2(\mathbf{y}), \ldots, g_K(\mathbf{y})\big) \tag{5}
$$

denote $K \in \mathbb{Z}_+$ such features. The predictor can then be expressed as

$$
\begin{aligned}
F(\bar{\mathbf{y}}) &= \Phi(g_1(\bar{\mathbf{y}}), g_2(\bar{\mathbf{y}}), \ldots, g_K(\bar{\mathbf{y}})) && (6)\\
&\approx \min_{\mathbf{x}\in\mathbb{R}^n} \left\{ C(\mathbf{x}) \mid A(\mathbf{x}) \le \mathbf{b},\ D(\mathbf{x}) \le \mathbf{q} - E(\bar{\mathbf{y}}),\ \mathbf{x} \in \mathcal{X} \right\}.
\end{aligned}
$$

Replacing the subproblem in (3)–(4) by the surrogate in (6) yields the following MESA-based approximation of the original model:

$$
\begin{aligned}
\min_{\pi_1,\ldots,\pi_K,\,\mathbf{y},\mathbf{z}} \quad & H(\mathbf{y}) + P(\mathbf{z}) + \Phi(\pi_1, \ldots, \pi_K) && (7)\\
\text{s.t.} \quad & \pi_j = g_j(\mathbf{y}), \qquad j = 1, \ldots, K, \\
& G(\mathbf{y}) + S(\mathbf{z}) \le \mathbf{r}, \\
& \pi_j \in \mathbb{R}, \qquad j = 1, \ldots, K, \\
& \mathbf{y} \in \mathcal{Y}, \qquad \mathbf{z} \in \mathcal{Z}. && (8)
\end{aligned}
$$

The feature equations and the function Φ collectively represent the trained predictor embedded in the optimization model.

When evaluating the predictor in (6) is substantially easier than repeatedly solving the original subproblem (3)–(4), the resulting approximation can yield substantial computational savings. At the same time, the quality of the approximated optimization model depends on the accuracy of

$F(\mathbf{y})$. In the ideal case in which

$$F(\mathbf{y}) = Q(\mathbf{y}), \qquad \forall\, \mathbf{y} \in \mathcal{Y}, \tag{9}$$

the surrogate reproduces the exact value function of the eliminated secondary decisions, and the MESA reformulation becomes equivalent to the original model with respect to the retained primary decisions.

Property Requirements of the Predictor Used in MESA

The predictor should balance approximation quality with MP representability and computational tractability. Linear and closed-form predictors are generally easier to embed into MP models than complex nonlinear models such as XGBoost or neural networks. Nevertheless, a broad range of nonlinear predictors—including decision trees, random forests, and neural networks—can be represented through mixed-integer formulations; see Gambella et al. (2021) and Maragno et al. (2023) for representative formulations and reviews.

Some modern MP solvers and associated modeling interfaces also support the integration of trained machine-learning models into optimization formulations. For example, Gurobi provides tools for embedding several commonly used predictive models into mathematical optimization workflows (Gurobi Optimization, LLC, 2023).

Predictive accuracy and formulation complexity must nevertheless be considered jointly. More complex predictive models may improve statistical fit, but their MP representations can introduce many auxiliary variables and constraints. The resulting increase in model size and formulation time may offset, or even exceed, the computational benefit obtained by replacing the original subproblem. Tree ensembles, for example, can require substantially larger embedded formulations than compact closed-form predictors.

Therefore, predictor selection is a central design decision in MESA. A suitable predictor should achieve a favorable trade-off between approximation accuracy and the size and complexity of the resulting optimization model. In the main implementation of this study, compact MP-representable predictors are adopted for A-CAM, while richer nonlinear alternatives are evaluated separately in the predictor-sensitivity analysis.

References

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